Machine shift report

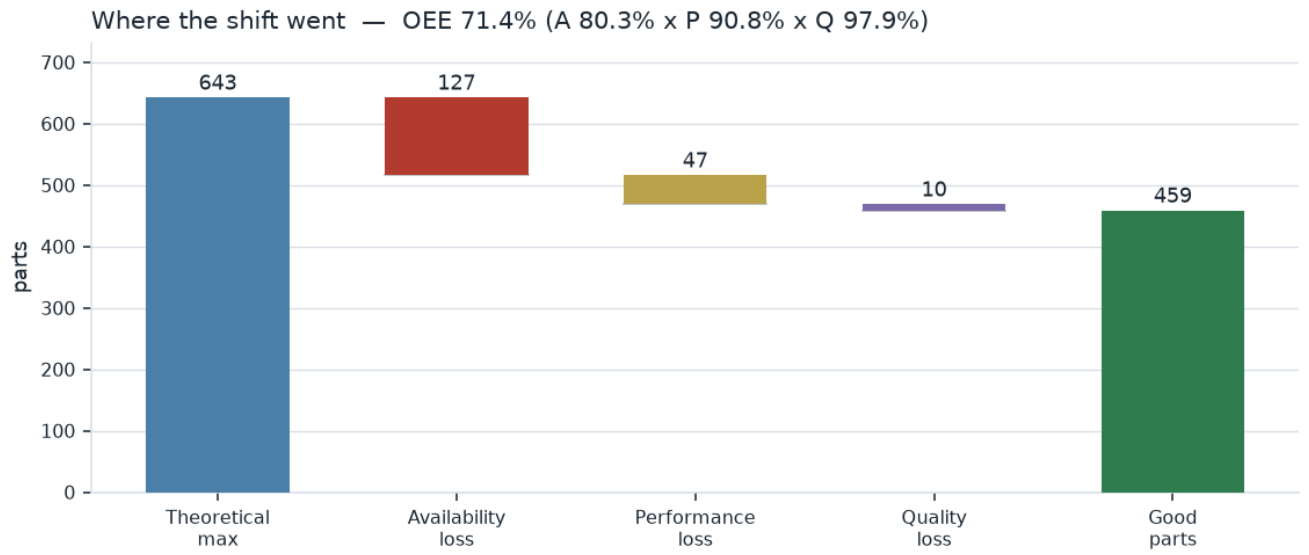
Collected over OPC UA from opc.tcp://127.0.0.1:4840/cnc/ · 1,541 samples · window 8h 00m · generated 2026-11-03 11:17

OEE 71.4%

Availability 80.3%, performance 90.8%, quality 97.9%, over 7h 30m of planned production time.

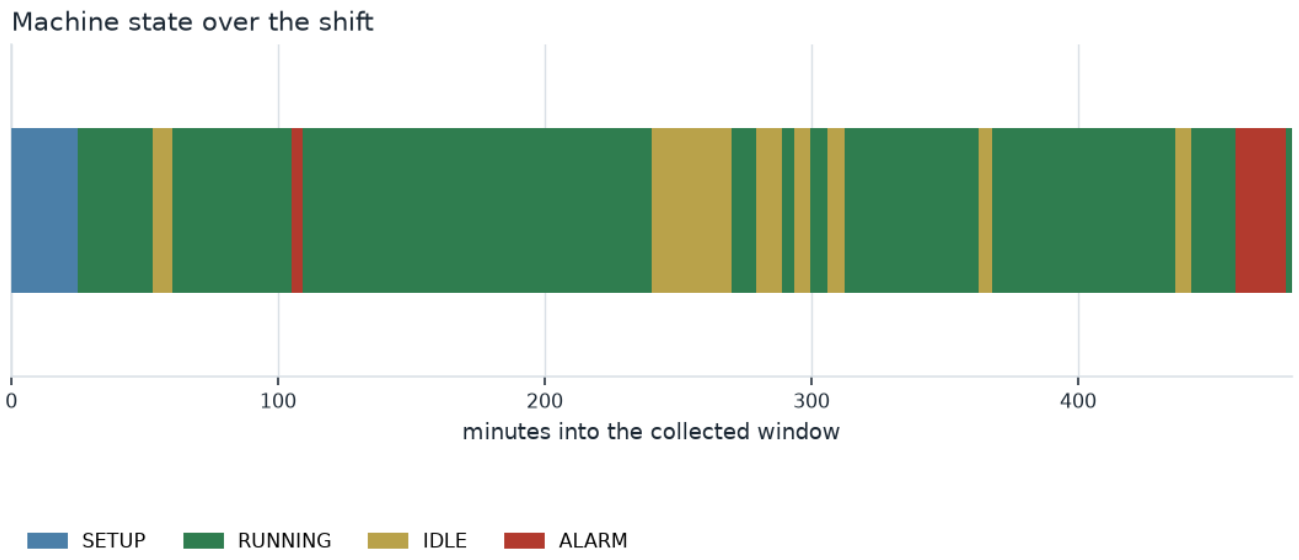
80.3%	90.8%	97.9%	459
availability	performance	quality	good parts

The three factors multiply, so a shift can lose as much to a slow cycle as to a stoppage: the waterfall shows which one actually cost the parts.



Where the time went

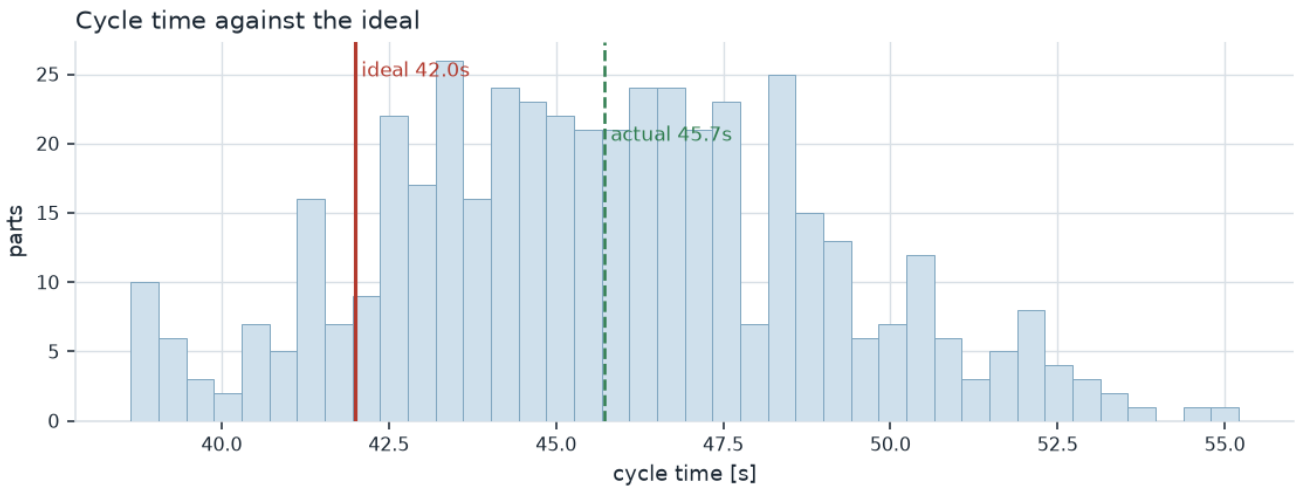
10 stops totalling 1h 58m. The longest was IDLE for 30 minutes. Many short stops and one long stop need different responses, which is why the timeline is here and not just the total.



bucket	duration	share of planned production time
RUNNING	6h 01m	80.3%
IDLE	1h 10m	15.7%
SETUP	0h 25m	5.6%
ALARM	0h 22m	5.1%
planned production time	7h 30m	100%
planned stops (excluded)	0h 30m	—

Rate and quality

Every part runs 3.7 s over the ideal cycle. Across 469 parts that is 0h 29m of the shift, or about 42 parts of lost output — the performance loss, expressed as something a process engineer can chase.



measure	value
parts produced	469
good	459
scrap	10 (2.13%)
ideal cycle time	42.0 s
actual mean cycle time	45.7 s
run time	6h 01m

Collection quality

Latency is the gap between the server's timestamp and local receipt. OEE here is computed from the source timestamp, so this latency does not bias the result — it only tells you how live a dashboard built on this feed would be.

measure	value
samples	1,541
dropped locally	0
median latency	107.0 ms
95th percentile	199.8 ms
worst	269.0 ms

OEE = availability × performance × quality. Availability is run time over planned production time; performance is ideal cycle time times total count over run time; quality is good count over total count. Planned stops are supplied from the shift plan, not inferred from the machine data, because a controller reports the same IDLE state for a planned break and for waiting on material. Counters are summed as positive deltas so a counter reset does not corrupt the total.